

Geometry
Unit 2
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

Complete the table for each given conditional statement.

If a shape is a triangle, then it has exactly three sides.			
	Relationship to Conditional	Statement	True or False
1.	Converse	If a shape has exactly three sides, then it is a triangle.	True
2.	Inverse	If a shape is not a triangle, then it does not have three sides.	True
3.	Contrapositive	If a shape does not have three sides, then it is not a triangle.	True
Write the conditional using "if and only if".			
4.	A shape is a triangle if and only if it has exactly three sides.		

Two angles are supplementary if their sum is 180° .			
	Relationship to Conditional	Statement	True or False
5.	Converse	If the sum of two angles is 180° , then they are supplementary.	True
6.	Inverse	If two angles are not supplementary, then their sum is not 180° .	True
7.	Contrapositive	If the sum of two angles is not 180° , then they are not supplementary.	True
Write the conditional using "if and only if".			
8.	Two angles are supplementary if and only if their sum is 180° .		

9. Write a conditional statement that is biconditional.

Answer will vary.

If a number is rational, then it can be written as a ratio of two integers.

Converse: If a number can be written as a ratio of two integers, it is rational.

Biconditional: A number is rational if and only if it can be written as a ratio of two integers.

10. Write a conditional statement that is **not** biconditional.

Answers will vary.

If you have a temperature over 115° , then you feel fine.

Converse: If you feel fine, then you will have a temperature of 115° . $\rightarrow$ False.

Biconditional: You will have a temperature of 115° if and only if you feel fine. $\rightarrow$ False